



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2011 Preliminary Course Assessment Task 2

General instructions

- Working time – 50 minutes.
- Commence each new question on a new page.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question. Marks may *not* be awarded for incomplete or untidy solutions.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 11M3A – Mr Weiss
- ☐ 11M3B – Mr Ireland
- ☐ 11M3C – Mr Berry
- ☐ 11M3D – Mrs Collins
- ☐ 11M3E – Mr Lam
- ☐ 11M3F – Mr Fletcher/Ms Everingham

NAME # BOOKLETS USED:

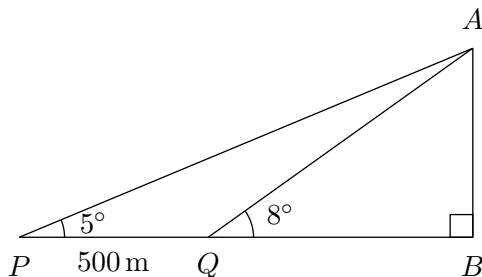
Marker's use only.

QUESTION	1	2	3	4	5	Total	%
MARKS	$\overline{11}$	$\overline{15}$	$\overline{8}$	$\overline{10}$	$\overline{5}$	$\overline{49}$	

Question 1 (11 Marks)	Commence a NEW page.	Marks
(a) Write $\frac{4\pi}{15}$ in degrees.		1
(b) Evaluate $\tan 1.5$ correct to two decimal places.		1
(c) Write down the exact value of		
i. $\cot 330^\circ$		1
ii. $\cos(-180^\circ)$		1
(d) Given $\tan \theta = \frac{3}{5}$ and $\cos \theta < 0$, find the exact value of $\operatorname{cosec} \theta$.		2
(e) Simplify $\sin(90^\circ - \theta) \times \tan(360^\circ - \theta)$.		2
(f) Given the function $y = 5 \cos 2x$		
i. State the amplitude		1
ii. State the period.		1
iii. Sketch the graph of the function for $0 \leq x \leq 180^\circ$.		1

Question 2 (15 Marks) Commence a NEW page. **Marks**

- (a) Steve observes a cliff AB from a point P . The angle of elevation to the top of the cliff is 5° . From Q , which is 500 m closer to the cliff, the angle of elevation to the top of the cliff is 8° .



- i. Calculate AQ to the nearest metre. **3**
- ii. Hence, find QB to the nearest 10 metres. **2**
- (b) Solve for $0^\circ \leq x \leq 360^\circ$:
- i. $\sin 2x = -\frac{1}{\sqrt{2}}$. **2**
- ii. $2 \sin^2 x = 3 \cos x$. **3**
- (c) Show that $\frac{\cot^2 \theta}{1 + \cot^2 \theta} = \cos^2 \theta$. **2**
- (d) In $\triangle ABC$, $AB = 25$ cm and $BC = 28$ cm. The area of $\triangle ABC$ is 340 cm^2 . **3**
- Find the value(s) of $\angle ABC$ to the nearest degree.

Question 3 (8 Marks)

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Marks

The coordinates of the point A , B and C are $(0, 2)$, $(4, 0)$ and $(6, -4)$ respectively.

- | | | |
|-----|---|----------|
| (a) | Find the length of AB . | 1 |
| (b) | Find the gradient of AB . | 1 |
| (c) | Find the equation of the line ℓ , drawn through C parallel to AB . | 2 |
| (d) | Find the coordinates of D , the point where ℓ intersects the x axis. | 1 |
| (e) | Find the perpendicular distance of the point A from the line ℓ . | 1 |
| (f) | Find the area of the quadrilateral $ABCD$. | 2 |

Question 4 (10 Marks)

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Marks

- | | | |
|-----|--|----------|
| (a) | Find the coordinates of the points $P(x, y)$ which divides the interval AB joining the points $A(-5, 11)$ and $B(7, 3)$ externally in the ratio $3 : 1$. | 2 |
| (b) | Find the equation of the line through the point of intersection of the lines $3x - 2y + 6 = 0$ and $2x - 3y + 1 = 0$ that is perpendicular to the line $x - y + 1 = 0$. | 4 |
| (c) | Sketch the region of the number plane where the following inequalities hold simultaneously: | 4 |

$$\begin{cases} y \geq x^2 - 4 \\ x + y < 2 \\ x \geq 0 \end{cases}$$

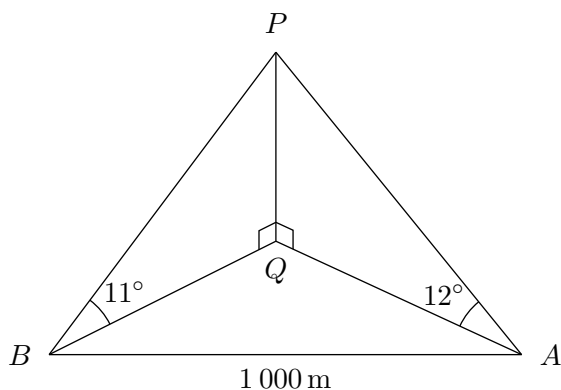
Question 5 (5 Marks)

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Marks

The angle of elevation of a tower PQ of height h metres from a point A due east of Q is 12° . From another point B , the bearing of the tower is 051°T and the angle of elevation is 11° .

$AB = 1\,000\text{ m}$ and AB is on the same level as the base Q .



(a) Show that $\angle AQB = 141^\circ$. **1**

(b) Show that **3**

$$h^2 = \frac{1\,000\,000}{\tan^2 78^\circ + \tan^2 79^\circ - 2 \tan 78^\circ \tan 79^\circ \cos 141^\circ}$$

(c) Find h , correct to the nearest metre. **1**

End of paper.

Question 1

(a) (1 mark)

$$\frac{4\pi}{15} = \frac{4 \times 180^\circ}{15} = 48^\circ$$

(b) (1 mark)

$$\tan 1.5 = 14.1$$

(c) i. (1 mark)

$$\cot 330^\circ = \frac{1}{\tan 330^\circ} = -\sqrt{3}$$

ii. (1 mark)

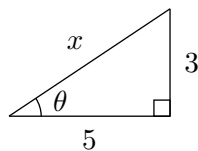
$$\cos(-180^\circ) = \cos 180^\circ = -1$$

(d) (2 marks)

✓ [1] for correct fraction.

✓ [1] for correct sign.

$$\tan \theta = \frac{3}{5} \quad \cos \theta < 0$$

Hence θ is in the 3rd quadrant.

$$x^2 = 3^2 + 5^2 = 34$$

$$\therefore x = \sqrt{34}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta} = -\frac{\sqrt{34}}{3}$$

(e) (2 marks)

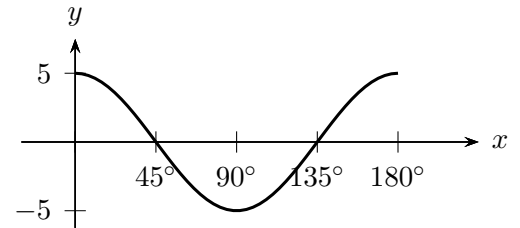
✓ [1] for $\cos \theta \times -\tan \theta$

✓ [1] for final answer

$$\begin{aligned} \sin(90^\circ - \theta) \times \tan(360^\circ - \theta) \\ &= \cos \theta \times -\tan \theta \\ &= \cos \theta \times \frac{-\sin \theta}{\cos \theta} \\ &= -\sin \theta \end{aligned}$$

(f) $y = 5 \cos 2x$ i. (1 mark) $-a = 5$.ii. (1 mark) $-T = \frac{360^\circ}{2} = 180^\circ$.

iii. (1 mark)

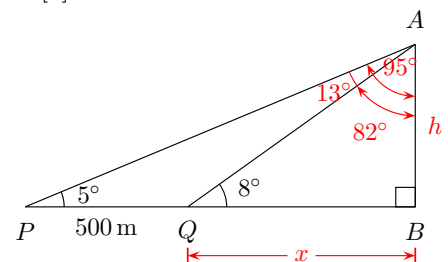
**Question 2**

(a) i. (3 marks)

✓ [1] for $\angle PAC = 3^\circ$

✓ [1] for applying the sine rule

✓ [1] for final answer



$$\angle QAB = 90^\circ - 8^\circ = 82^\circ$$

$$\angle PAB = 90^\circ - 5^\circ = 85^\circ$$

$$\therefore \angle PAQ = 85^\circ - 82^\circ = 3^\circ$$

Applying the sine rule on $\triangle PAQ$,

$$\begin{aligned} \frac{AQ}{\sin 5^\circ} &= \frac{500}{\sin 3^\circ} \\ AQ &= \frac{500 \sin 5^\circ}{\sin 3^\circ} = 833 \text{ m} \end{aligned}$$

ii. (2 marks)

✓ [1] for $\frac{QB}{AQ} = \cos 8^\circ$

✓ [1] for final answer

$$\begin{aligned} \frac{QB}{AQ} &= \cos 8^\circ \\ QB &= AQ \cos 8^\circ \\ &= \frac{500 \sin 5^\circ \cos 8^\circ}{\sin 3^\circ} = 820 \text{ m} \end{aligned}$$

(b) i. (2 marks)

✓ [1] for solutions in $2x$ ✓ [1] for final solutions in x .

$$\sin 2x = -\frac{1}{\sqrt{2}} \quad 0^\circ \leq 2x \leq 720^\circ$$

$$2x = 225^\circ, 315^\circ, 360^\circ + 225^\circ, 360^\circ + 315^\circ \\ = 225^\circ, 315^\circ, 585^\circ, 675^\circ$$

$$\therefore x = 112.5^\circ, 157.5^\circ, 292.5^\circ, 337.5^\circ$$

ii. (3 marks)

✓ [1] for $2 - 2\cos x = 3\cos x$.✓ [1] for $2\cos x = 1$ or $\cos x = -2$.

$$2\sin^2 x = 3\cos x$$

$$2(1 - \cos^2 x) = 3\cos x$$

$$2 - 2\cos^2 x = 3\cos x$$

$$2\cos^2 x + 3\cos x - 2 = 0$$

Let $m = \cos x$

$$2m^2 + 3m - 2 = 0$$

$$(2m - 1)(m + 2) = 0$$

$$\therefore m = \cos x = \frac{1}{2}, -2$$

However, $|\cos x| \leq 1$. $\therefore \cos x = \frac{1}{2}$
only

$$x = 60^\circ, 300^\circ$$

(c) (2 marks)

✓ [1] for $\frac{\cot^2 \theta}{\operatorname{cosec}^2 \theta}$

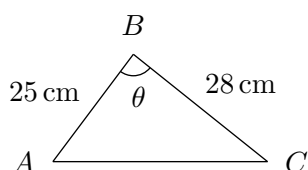
✓ [1] for final answer

$$\frac{\cot^2 \theta}{1 + \cot^2 \theta} = \frac{\frac{\cos^2 \theta}{\sin^2 \theta}}{\operatorname{cosec}^2 \theta} = \frac{\frac{\cos^2 \theta}{\sin^2 \theta}}{\frac{1}{\sin^2 \theta}} = \cos^2 \theta$$

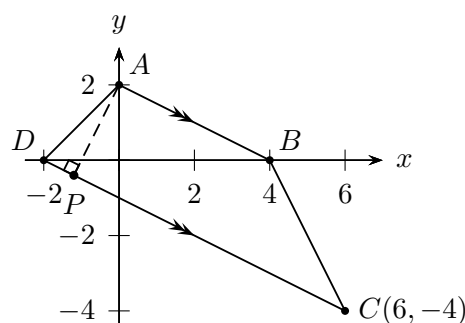
(d) (3 marks)

✓ [1] for applying area of triangle formula via sine

✓ [2] for final solutions.

**Question 3**

(a) (1 mark)



$$AB = \sqrt{4^2 + 2^2} = \sqrt{20} = 2\sqrt{5}$$

(b) (1 mark) $m_{AB} = \frac{-2}{4} = -\frac{1}{2}$

(c) (2 marks)

✓ [1] for applying the point gradient formula

✓ [1] for final answer

$$m_\ell = m_{AB} = -\frac{1}{2}$$

Applying the point gradient formula

$$y + 4 = -\frac{1}{2}(x - 6)$$

$$2y + 8 = -x + 6$$

$$x + 2y + 2 = 0 \quad \left(y = -\frac{1}{2}x - 1 \right)$$

(d) (1 mark)

$$x + 2y + 2 \Big|_{y=0} = 0$$

$$x + 2 = 0 \Rightarrow x = -2$$

$$\therefore D(-2, 0)$$

(e) (1 mark)

$$d_{\perp} = \frac{|1(0) + 2(2) + 2|}{\sqrt{1^2 + 2^2}} = \frac{6}{\sqrt{5}}$$

(f) (2 marks)

$$CD = \sqrt{8^2 + 4^2} = \sqrt{80} = 4\sqrt{5}$$

As $ABCD$ is a trapezium,

$$\begin{aligned} A &= \frac{1}{2}h(a+b) \\ &= \frac{1}{2} \times \frac{6}{\sqrt{5}} (2\sqrt{5} + 4\sqrt{5}) \\ &= \frac{3}{\cancel{\sqrt{5}}} \times 6\cancel{\sqrt{5}} = 18 \end{aligned}$$

Question 4

(a) (2 marks)

$$\begin{aligned} P &= \left(\frac{3(7) + (-5)(-1)}{3-1}, \frac{3(3) + 11(-1)}{3-1} \right) \\ &= (13, -1) \end{aligned}$$

(b) (4 marks)

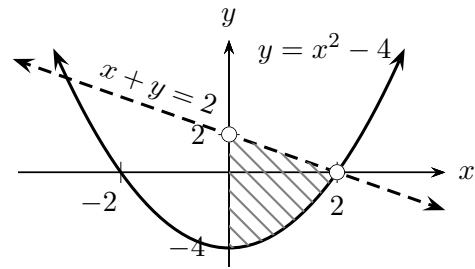
- ✓ [1] for placing two equations in k form
- ✓ [1] for $\frac{3+2k}{2+3k} = -1$
- ✓ [1] for $k = -1$
- ✓ [1] for $x + y + 5 = 0$

$$\begin{aligned} 3x - 2y + 6 + k(2x - 3y + 1) &= 0 \\ (3 + 2k)x + (-2 - 3k)y + (k + 6) &= 0 \end{aligned}$$

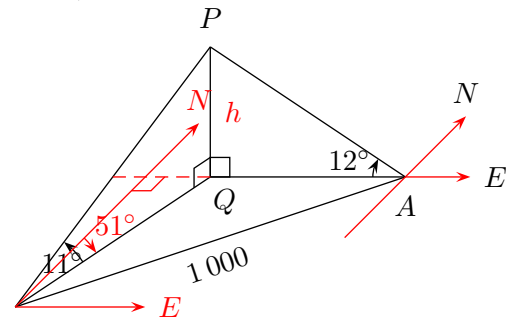
The gradient of the line $x - y + 1 = 0$ is $m = 1$. Hence required gradient is $m = -1$.

$$\begin{aligned} \therefore \frac{3+2k}{2+3k} &= -1 \\ 3+2k &= -2-3k \\ -5k &= 5 \Rightarrow k = -1 \\ \therefore (3-2)x + (-2+3)y + (-1+6) &= 0 \\ x + y + 5 &= 0 \end{aligned}$$

(c) (4 marks)

**Question 5**

(a) (1 mark) Redraw the diagram:



$\angle AQB = 90^\circ + 51^\circ = 141^\circ$
(exterior $\angle$ of $\triangle$ is the sum of the interior opposite angles.)

(b) (3 marks)

- ✓ [1] for $BQ = h \tan 78^\circ$ (or for AQ)
- ✓ [1] for applying the cosine rule
- ✓ [1] for final answer

In $\triangle BQP$,

$$\begin{aligned} \frac{BQ}{h} &= \tan(90^\circ - 11^\circ) = \tan 79^\circ \\ \therefore BQ &= h \tan 79^\circ \end{aligned}$$

Similarly in $\triangle AQP$,

$$\therefore AQ = h \tan 78^\circ$$

Applying the cosine rule in $\triangle ABQ$,

$$\begin{aligned} 1000^2 &= (h \tan 79^\circ)^2 + (h \tan 78^\circ)^2 \\ &\quad - 2(h \tan 78^\circ)(h \tan 79^\circ) \cos 141^\circ \\ &= h^2 (\tan^2 79^\circ + \tan^2 78^\circ \\ &\quad - 2 \tan 78^\circ \tan 79^\circ \cos 141^\circ) \\ \therefore h^2 &= \frac{1\,000\,000}{\tan^2 79^\circ + \tan^2 78^\circ - 2 \tan 78^\circ \tan 79^\circ \cos 141^\circ} \end{aligned}$$

(c) (1 mark)

$$h = 108 \text{ m}$$